Notes version: 11 Nov 19:02 IST (if this time is old, reload the page)

HISTORY · National Movement IV: Cripps, Quit India, INA, Cabinet Mission

UNDERSTAND IT FIRST

What is it? The **Second World War (1939–45)** changed everything. Britain dragged India into the war **without asking Indians**, needed Indian help, and came out of the war **weak**. Indians now demanded **freedom, not reforms**: through the **Quit India Movement (1942)**, the last great mass struggle, and through **armed struggle** by Netaji's **Indian National Army**. Britain's offers (**Cripps 1942**, **Cabinet Mission 1946**) show its move from **delay to withdrawal**, while the **Muslim League's** demand for **Pakistan** grew. Learn it as **war and the first offers → Cripps → Quit India → INA → 1945–46 (Simla, elections, mutinies) → Cabinet Mission → road to 1947**.

From the war to the Cabinet Mission

1. The war years, 1939–41

- **3 September 1939:** Viceroy **Linlithgow** declared India at war **without consulting** Indian leaders; the **Congress ministries resigned** (October–November 1939); the Muslim League celebrated a **'Day of Deliverance' (22 December 1939)**.
- **Lahore Resolution (23 March 1940):** the Muslim League demanded **separate 'independent States'** for Muslim-majority areas (the **'Pakistan Resolution'**, moved by **Fazlul Huq**). **August Offer (8 August 1940): dominion status** as the aim (no date), more Indians in the Viceroy's Council, and a **veto for minorities** on any future constitution; rejected. **Individual Satyagraha (October 1940):** chosen individuals protested against the war; first **Vinoba Bhave**, second **Jawaharlal Nehru**.

2. The Cripps Mission (March 1942)

- **Why: Japan** was advancing (Singapore fell, **Rangoon fell, March 1942**) and Britain needed Indian support; the USA and China also pressed Britain. **Sir Stafford Cripps** offered: **dominion status after the war**; a **constitution-making body** elected by provincial assemblies and nominated by princes; **any province could stay out** and form its own union; defence to remain with Britain during the war.
- **Result: rejected by both** the Congress (no real power now; the opt-out clause threatened unity) and the League (no clear Pakistan). Gandhi called it a **'post-dated cheque'** (on a crashing bank).

3. The Quit India Movement (1942)

Point	Details
Resolution	Congress Working Committee at **Wardha (14 July 1942)**; the **AICC at Gowalia Tank, Bombay** passed the **Quit India resolution on 8 August 1942**; Gandhi's call: **'Do or Die'** ('Karo ya Maro').
Course	Early on **9 August 1942** ('Operation Zero Hour') all top leaders were **arrested**; **Aruna Asaf Ali** hoisted the flag at Gowalia Tank. Leaderless, people **attacked railways, telegraph lines and**

	police stations; underground leaders: **Jayaprakash Narayan, Ram Manohar Lohia, Aruna Asaf Ali**; **Usha Mehta** ran a secret **Congress Radio**.
Parallel governments	**Ballia** (UP, August 1942; **Chittu Pandey**); **Tamluk** (Midnapore, Bengal; **Jatiya Sarkar**, 1942–44; **Matangini Hazra** was shot leading a procession); **Satara** (Maharashtra; **Prati Sarkar**, 1943–45; **Nana Patil**).
Who stayed away	The **Muslim League**, the **Communist Party** (after the USSR joined the Allies, it called it a 'People's War'), the **Hindu Mahasabha**, and many **princes**; the **Depressed Classes** leadership did not join.
After	Crushed by **1943**; Gandhi's **21-day fast (February 1943)** in the **Aga Khan Palace**; **Kasturba died there (February 1944)**; Gandhi released (**May 1944**); the **C. Rajagopalachari formula (1944)** and **Gandhi–Jinnah talks (September 1944)** failed. **Significance:** showed India's **determination**; independence became **certain after the war**.
Andhra	Strikes and protests spread across the Andhra districts; crowds attacked the railway station at **Tenali** (August 1942), where police firing killed several people; incidents also at **Bhimavaram, Chirala** and other towns; students and women joined.

4. Subhas Chandra Bose and the INA

- **Bose's path:** Congress president at **Haripura (1938)** and **Tripuri (1939)**; differences with Gandhi made him resign and form the **Forward Bloc (1939)**; escaped house arrest in **Calcutta (January 1941)** to **Germany** (Free India Centre, **Azad Hind Radio**, an Indian Legion; called **'Netaji'**); reached **Singapore (July 1943)** by submarine.
- **INA:** first formed by **Captain Mohan Singh** (1942) from Indian prisoners of war after the fall of **Singapore**; **Rash Behari Bose** led the **Indian Independence League**; Netaji took command (1943) and proclaimed the **Provisional Government of Free India (Azad Hind) at Singapore on 21 October 1943**; slogans **'Delhi Chalo'**, **'Give me blood, and I will give you freedom'**, **'Jai Hind'**; brigades named **Gandhi, Nehru, Azad**; the women's **Rani of Jhansi Regiment** (**Lakshmi Swaminathan / Sahgal**); Netaji hoisted the flag at **Port Blair (30 December 1943)**; INA flag at **Moirang, Manipur (April 1944)**; the **Imphal–Kohima** campaign failed (1944); Netaji reportedly died in an **air crash at Taipei (18 August 1945)**.
- **INA trials (Red Fort, November 1945):** officers **Shah Nawaz Khan, Prem Sahgal and Gurbaksh Singh Dhillon** (a Muslim, a Hindu and a Sikh) were defended by **Bhulabhai Desai**, Tej Bahadur Sapru, **Jawaharlal Nehru** and Asaf Ali; the whole country protested and the sentences were remitted. **Royal Indian Navy mutiny (Bombay, 18 February 1946, HMIS Talwar)**: showed that Britain could **no longer trust Indian forces**.

5. 1945–46: Simla, Cabinet Mission, Interim Government

Event	Facts
Wavell Plan and Simla	An all-Indian Executive Council (except the Viceroy and the C-in-C); **failed** because **Jinnah insisted** the League alone would nominate all Muslim members.

Conference (June 1945)	
Elections (1945–46)	The Congress won most **general** seats; the League won most **Muslim** seats, strengthening its claim to Pakistan. In Britain, **Labour (Attlee)** had come to power (July 1945).
Cabinet Mission (March–June 1946)	**Pethick-Lawrence, Stafford Cripps, A.V. Alexander**; plan of **16 May 1946**: **rejected a separate Pakistan**; a **weak Union** (defence, foreign affairs, communications); provinces in **three groups**: **A** (Hindu-majority provinces), **B** (Punjab, NWFP, Sindh), **C** (Bengal, Assam); a **Constituent Assembly of 389** (292 from provinces, 4 from chief commissioners' provinces, 93 from princely States), elected **indirectly** by provincial assemblies; an **interim government**.
Direct Action and after	The League first accepted, then **withdrew** and called **'Direct Action Day' (16 August 1946)**: terrible **riots in Calcutta**, then Noakhali and Bihar. **Interim Government** formed **2 September 1946** with **Nehru** as Vice-President of the Executive Council; the League joined in **October 1946** (Liaquat Ali Khan, Finance) but obstructed. The **Constituent Assembly met on 9 December 1946** (League boycotted). **Attlee's statement (20 February 1947):** Britain would leave **by June 1948**.

6. Impact

Area	Points
Politics	Britain accepted that it must **transfer power**; but the **League–Congress gap** widened and **Partition** became likely.
Society	**Youth, students, women and peasants** led Quit India; the INA showed **unity across religions**; but 1946 brought **communal violence**.
Economy	War brought **inflation, shortages** and the **Bengal Famine (1943)**, in which about **three million** died; Britain became **indebted to India** (sterling balances).
Religion	The **two-nation** argument hardened (Lahore 1940, Direct Action 1946).

EXAM TRAP

3 Sept 1939 war; ministries resign; **Day of Deliverance 22 Dec 1939**; **Lahore Resolution 23 Mar 1940**; **August Offer 8 Aug 1940**; **Individual Satyagraha Oct 1940** (Vinoba first); **Cripps Mar 1942** ('post-dated cheque'); **Quit India 8 Aug 1942** (Gowalia Tank, Do or Die); **9 Aug** arrests; Aruna Asaf Ali, JP, Lohia, Usha Mehta; **Ballia, Tamluk, Satara**; **Tenali** (Andhra); **Bengal Famine 1943**; **Kasturba d. Feb 1944**; **CR formula 1944**; **INA**: Mohan Singh 1942; **Azad Hind Govt 21 Oct 1943**; **Port Blair 30 Dec 1943**; **Moirang 1944**; **Red Fort trials Nov 1945**; **RIN mutiny 18 Feb 1946**; **Simla June 1945**; **Cabinet Mission plan 16 May 1946** (A, B, C; 389); **Direct Action 16 Aug 1946**; **Interim Govt 2 Sept 1946**; **CA 9 Dec 1946**; **Attlee 20 Feb 1947**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about the Cripps Mission (1942):

1. It offered dominion status to India after the war.

2. It allowed any province to stay out of the proposed Indian Union.

3. Gandhi called its offer a 'post-dated cheque'.

4. The Muslim League accepted its proposals.

Which of the above are correct?

A. 1 and 2 only

B. 1, 2 and 3 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: B. Both the Congress and the Muslim League rejected the Cripps proposals.

PRACTICE

Match List-I (place or activity during Quit India) with List-II (person or body):

List-I (place or activity during Quit India)	List-II (person or body)
A. Ballia	1. Usha Mehta
B. Tamluk	2. Prati Sarkar under Nana Patil
C. Satara	3. Jatiya Sarkar in Midnapore
D. Secret Congress Radio	4. Chittu Pandey

(1) A → 3 B → 4 C → 2 D → 1

(2) A → 4 B → 3 C → 2 D → 1

(3) A → 4 B → 2 C → 3 D → 1

(4) A → 4 B → 3 C → 1 D → 2

Answer: (2) Ballia: Chittu Pandey; Tamluk: Jatiya Sarkar; Satara: Prati Sarkar; radio: Usha Mehta.

PRACTICE

Which one of the following statements about the Indian National Army is NOT correct?

A. The Provisional Government of Azad Hind was proclaimed at Singapore in October 1943.

B. The first INA was raised by Subhas Chandra Bose in Germany in 1941.

C. The Rani of Jhansi Regiment was led by Lakshmi Swaminathan.

D. Bhulabhai Desai defended the INA officers at the Red Fort trials.

Answer: B. The first INA was raised by Captain Mohan Singh in 1942 from prisoners of war after the fall of Singapore.

PRACTICE

Assertion (A): The Quit India resolution was passed by the AICC at Bombay on 8 August 1942.
Reason (R): The Muslim League and the Communist Party actively supported the Quit India Movement.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: C. A is true. R is false: the League and the Communists stayed away from the movement.

PRACTICE

Arrange the following in chronological order:
1. Simla Conference
2. Lahore Resolution of the Muslim League
3. Quit India Resolution
4. Direct Action Day

A. 2, 1, 3, 4

B. 3, 2, 1, 4

C. 2, 3, 1, 4

D. 2, 3, 4, 1

Answer: C. Lahore 1940, Quit India 1942, Simla 1945, Direct Action 1946.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write 1939–41 events and the Cripps proposals. 2. Write Quit India: resolution, course, parallel governments, Andhra, who stayed away. 3. Write the INA and 1945–46 events up to Attlee's statement.

POLITY · Political parties and the party system; recognition of parties

UNDERSTAND IT FIRST

What is it? A **political party** is an organised group of people with **shared ideas** who seek to **win elections and form a government**. Parties **contest elections, form governments or act as the opposition, make policies, shape public opinion** and **link people with the government**. The **party system** describes **how many parties** effectively compete: **one-party** (China), **two-party** (USA, broadly UK) or **multi-party** (India). India's Constitution originally **did not mention parties**; their legal base comes from the **freedom to form associations (Art 19(1)(c))**, the **Representation of the People Act**, the **Election Commission's Symbols Order** and the **Tenth Schedule**.

1. India's party system: four phases

Phase	Features
1952–1967	**One-party dominance**: the Congress won at the Centre and in almost all States; **Rajni Kothari** called it the **'Congress system'** (opposition worked as pressure groups inside and outside the party).
1967–1989	**1967**: the Congress lost power in several States (coalitions, 'Aaya Ram Gaya Ram' defections); **1977**: the **Janata Party** formed the **first non-Congress government** at the Centre after the Emergency; Congress dominance returned in 1980 and 1984.
1989–2014	**Coalition era**: no single party won a Lok Sabha majority; **regional parties** became powerful at the Centre (National Front, United Front, NDA, UPA).
2014 onwards	In **2014**, a single party (the BJP) won a Lok Sabha majority for the **first time in 30 years**, governing with allies; regional parties remain strong in many States.

2. Registration and recognition

- **Registration:** under **Section 29A of the Representation of the People Act, 1951**, a party registers with the **Election Commission**, pledging loyalty to the **Constitution, socialism, secularism and democracy** and the **sovereignty, unity and integrity** of India. Most registered parties remain **'registered unrecognised'**.
- **Recognition:** the EC recognises **National** and **State** parties under the **Election Symbols (Reservation and Allotment) Order, 1968**, based on **votes and seats** in the **last general election** (criteria in the diagram). The EC **reviews** status after elections; for example, in **2023** it listed **six national parties**: **BJP, INC, BSP, CPI(M), AAP and NPP** (the National People's Party).

Recognition criteria

3. Issues and reforms

Issue	Details
Funding	Parties must report donations above **₹20,000** (Sec 29C); the **Electoral Bonds Scheme (2018)** allowed anonymous donations, but the Supreme Court **struck it down on 15 February 2024** (ADR case) as violating voters' **right to information**; the **Indrajit Gupta Committee (1998)** suggested **partial State funding** of elections.
Criminalisation	**Lily Thomas (2013)**: MPs and MLAs are **disqualified immediately** on conviction with a sentence of **2 years or more**; the Supreme Court (**2018, 2020**) told parties to **publish candidates' criminal records** and explain why they chose them.
Other issues	Little **internal democracy** (Law Commission **170th Report, 1999**); **dynastic** politics; **defections** (Tenth Schedule, **52nd Amendment, 1985**, the Constitution's first mention of parties; **91st Amendment (2003)** removed the 'split' exception); whether parties come under **RTI** (the CIC said yes in 2013; still disputed).

EXAM TRAP

Functions; **one / two / multi-party**; India: **1952–67 Congress system (Rajni Kothari)**, **1967** coalitions, **1977 Janata**, **1989–2014 coalitions**, **2014** single-party majority; **Art 19(1)(c)**; **RPA Sec 29A** registration, **29C** donations over ₹20,000; **Symbols Order 1968**; **State party**: 6% + 2 seats, 6% + 1 LS, 3% seats (min 3), 1 per 25 LS, or 8% votes; **National party**: 6% in 4 States + 4 LS, 2% LS seats from 3 States, or State party in 4 States; **6 national parties (2023)**; benefits: symbol, airtime, **40 star campaigners**; **bonds struck down 15 Feb 2024**; **Indrajit Gupta 1998**; **Lily Thomas 2013**; **52nd (1985)**, **91st (2003)**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:
1. The Tenth Schedule, added by the 52nd Amendment, was the first mention of political parties in the Constitution.
2. Political parties register with the Election Commission under Section 29A of the Representation of the People Act, 1951.
3. Parties are recognised as national or State parties under the Election Symbols (Reservation and Allotment) Order, 1968.
4. The Constitution as originally adopted contained a separate chapter on political parties.
Which of the above are correct?

A. 1 and 2 only

B. 2, 3 and 4 only

C. 1, 2 and 3 only

D. 1, 2, 3 and 4

Answer: C. The original Constitution did not mention political parties at all.

PRACTICE

Match List-I (period) with List-II (feature of India's party system):

List-I (period)	List-II (feature of India's party system)
A. 1952–1967	1. A single party won a Lok Sabha majority for the first time in 30 years
B. 1977	2. Era of coalition governments
C. 1989–2014	3. First non-Congress government at the Centre
D. 2014	4. One-party dominance, the 'Congress system'

(1) A → 4 B → 3 C → 2 D → 1

(2) A → 3 B → 4 C → 2 D → 1

(3) A → 4 B → 2 C → 3 D → 1

(4) A → 4 B → 3 C → 1 D → 2

Answer: (1) 1952–67: Congress system; 1977: Janata; 1989–2014: coalitions; 2014: single-party majority.

PRACTICE

Which one of the following statements is NOT correct?

A. A party recognised as a State party in four States qualifies as a national party.

B. A recognised party can name up to 40 star campaigners.

C. The Supreme Court struck down the Electoral Bonds Scheme in 2024.

D. To be a national party, a party must win at least 6% of votes in every State.

Answer: D. The 6% rule applies in at least four States (with 4 Lok Sabha seats), not every State.

PRACTICE

Assertion (A): India's party system became more competitive after 1967.

Reason (R): In the 1967 elections the Congress lost power in several States, and non-Congress and

regional parties formed governments.

Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. The 1967 losses ended one-party dominance in the States.

PRACTICE

In a State Assembly general election, a registered party wins 8.5% of the total valid votes in the State but no seats. Under the Symbols Order, what is its position?

A. It cannot be recognised because it won no seat

B. It qualifies as a national party

C. It qualifies as a State party

D. It must wait for two more elections

Answer: C. Securing 8% of the total valid votes in the State is enough by itself for State-party recognition.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the functions of parties, types of party systems and India's four phases. 2. Write registration (29A) and the State and national party criteria with benefits. 3. Write funding, bonds, criminalisation and other reforms.

SCIENCE · Nuclear technology: reactors, power plants, radioisotopes

UNDERSTAND IT FIRST

What is it? The **nucleus** of an atom holds enormous energy. In **fission**, a heavy nucleus (like **uranium-235**) is hit by a **neutron** and **splits** into two smaller nuclei, releasing **heat** and **2–3 more neutrons**, which split more nuclei: a **chain reaction**. A **nuclear reactor** keeps this chain reaction **steady and controlled**; the heat makes **steam**, which turns a **turbine** to produce electricity, with **no carbon dioxide**. (In **fusion**, light nuclei such as hydrogen **join**, as in the Sun; it is still experimental, e.g. **ITER**, where India is a partner.) Nuclear science also gives **radioisotopes** for **medicine, farming and industry**.

Parts of a nuclear reactor

1. Reactor types and the three-stage programme

- **Types: PHWR** (pressurised heavy water reactor): **natural uranium** fuel with **heavy water** as moderator and coolant; India's mainstay. **LWR/PWR**: **enriched uranium** with ordinary water (e.g. the Russian **VVER** at **Kudankulam**). **Fast breeder reactor (FBR)**: no moderator, **liquid sodium** coolant; **produces more fissile fuel than it uses**. **AHWR**: a proposed **thorium**-based design.
- **Why three stages?** India has **little uranium** but some of the **world's largest thorium** reserves (in **monazite beach sands** of Kerala, Odisha, Tamil Nadu and **Andhra Pradesh**). Thorium is **not fissile**; it must first be turned into **U-233** in a reactor. So **Homi J. Bhabha** planned three stages (1954). The **500 MW Prototype Fast Breeder Reactor (PFBR), Kalpakkam** began **fuel loading in 2024**.

Three-stage programme

2. Institutions, plants and policy

Topic	Facts
Institutions	**Atomic Energy Commission (1948)**, chair **Homi Bhabha**; **Department of Atomic Energy (1954)**, under the **PM**; **BARC** (Trombay, Mumbai); **IGCAR** (Kalpakkam); **NPCIL (1987)** runs power plants; **BHAVINI (2003)** for fast breeders; **AERB (1983)** is the safety regulator; **Uranium Corporation of India** (mines at **Jaduguda**, Jharkhand, and **Tummalapalle, Kadapa (YSR) district**, AP); **Nuclear Fuel Complex, Hyderabad**; **IREL** processes beach-sand minerals.
Plants	**Apsara (1956)**, Trombay: **Asia's first research reactor**; **Tarapur (1969)**: first nuclear **power** station; others at **Rawatbhatta, Kalpakkam, Narora, Kakrapar** (the first indigenous **700 MW PHWR**, 2021), **Kaiga** and **Kudankulam** (largest, Russian VVERs); **Kovvada (Srikakulam, AP)** is a **planned** site. Installed nuclear capacity is about **8,000+ MW** (about 3% of electricity); the target is **100 GW by 2047** with **small modular reactors** (Budget 2025).
Tests, doctrine, agreements	**Pokhran-I (18 May 1974, 'Smiling Buddha')**; **Pokhran-II (11 and 13 May 1998, 'Operation Shakti')**: **11 May** is **National Technology Day**; doctrine of **'No First Use'** and **credible minimum deterrence (2003)**; India has **not signed the NPT or the CTBT**; **India–US civil nuclear deal (2008)** with an **NSG waiver**; **Civil Liability for Nuclear Damage Act, 2010**.

3. Radioisotopes and safety

Use	Examples
Medicine	**Cobalt-60** for cancer radiotherapy; **Iodine-131** to treat **thyroid** disorders and cancer; **Technetium-99m** for imaging scans; **sterilising** syringes and surgical items.
Agriculture and food	**Radiation-induced mutation** gives new crop varieties (BARC's groundnut, pulses); **food irradiation** stops sprouting and extends **shelf life** (onions, potatoes, mangoes); **sterile insect technique** for pest control.

Industry and research	**Radiography** to find cracks in welds and pipes; **thickness gauges**; **carbon-14 dating** of ancient objects.
Safety concerns	Accidents at **Chernobyl (1986)** and **Fukushima (2011)**; long-lived **radioactive waste**; high **cost** and **local opposition** (e.g. Kudankulam protests).

EXAM TRAP

Fission (U-235, chain reaction) vs **fusion** (ITER); reactor parts: **fuel, moderator, control rods, coolant, shielding**; **PHWR** (natural U, heavy water), **PWR/VVER**, **FBR** (sodium), **AHWR**; **three stages (Bhabha 1954)**; **PFBR Kalpakkam 500 MW** (fuel loading 2024); **AEC 1948**, **DAE 1954**, **BARC**, **NPCIL 1987**, **AERB 1983**; **Tummalapalle (AP)**, **NFC Hyderabad**, monazite sands; **Apsara 1956**, **Tarapur 1969**, **Kakrapar 700 MW 2021**, **Kudankulam**, **Kovvada** (planned); **~8,000+ MW**, **100 GW by 2047**; **Pokhran 1974, 1998** (11 May); **NFU 2003**; no NPT, CTBT; **2008 deal**; **2010 liability Act**; **Co-60, I-131, Tc-99m**; food irradiation.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about India's three-stage nuclear programme:
1. The first stage uses pressurised heavy water reactors fuelled by natural uranium.
2. The second stage uses fast breeder reactors fuelled by plutonium.
3. The third stage aims to use thorium through uranium-233.
4. India has much larger reserves of uranium than of thorium.
Which of the above are correct?

A. 1 and 2 only

B. 2, 3 and 4 only

C. 1, 2, 3 and 4

D. 1, 2 and 3 only

Answer: D. India has limited uranium but very large thorium reserves; that is why the programme ends with thorium.

PRACTICE

Match List-I (part of a reactor) with List-II (function):

List-I (part of a reactor)	List-II (function)
A. Moderator	**1.** Stops radiation from escaping
B. Control rods	**2.** Carries away heat

C. Coolant	**3.** Absorbs neutrons to control the reaction
D. Shielding	**4.** Slows down neutrons

(1) A → 3 B → 4 C → 2 D → 1

(2) A → 4 B → 2 C → 3 D → 1

(3) A → 4 B → 3 C → 2 D → 1

(4) A → 4 B → 3 C → 1 D → 2

Answer: (3) Moderator slows; control rods absorb; coolant carries heat; shielding stops radiation.

PRACTICE

Which one of the following statements is NOT correct?

A. Apsara (1956) was Asia's first research reactor.

B. Kudankulam uses Russian-designed VVER reactors.

C. Pokhran-II tests were carried out in 1974.

D. Tarapur was India's first nuclear power station.

Answer: C. Pokhran-I was in 1974; Pokhran-II was on 11 and 13 May 1998.

PRACTICE

Assertion (A): Food irradiation extends the shelf life of onions and potatoes.

Reason (R): India conducted its first nuclear test at Pokhran in 1974.

Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: B. Both are true but unrelated: irradiation stops sprouting and kills microbes.

PRACTICE

A hospital treats a patient's overactive thyroid gland with a radioisotope that the thyroid absorbs. Which radioisotope is used?

A. Cobalt-60

B. Carbon-14

C. Iodine-131

D. Uranium-235

Answer: C. The thyroid takes up iodine, so Iodine-131 is used for thyroid treatment.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write fission and fusion, the five reactor parts and the reactor types. 2. Write the three-stage programme, institutions and plants. 3. Write tests, doctrine, agreements, radioisotope uses and AP links.

NOTEBOOK PAGE · Day 22: write this from memory

Close the notes. Write these headings, fill every blank, then correct in a different colour.

HISTORY: National Movement IV

Area	Write
1939–1941	War declared, ministries resign, Day of Deliverance, Lahore Resolution, August Offer, Individual Satyagraha
Cripps and Quit India	Cripps proposals; Wardha, Bombay 8 Aug 1942, Do or Die, underground leaders, parallel governments, Andhra events, who stayed away
INA	Mohan Singh, Rash Behari Bose, Netaji's escape, Azad Hind Government, Port Blair, Imphal–Kohima, Red Fort trials, RIN mutiny
1945–1947	Wavell Plan, Simla, Cabinet Mission (groups, Assembly), Direct Action, Interim Government, Attlee 1947

POLITY: Political parties

- Functions; party systems; four phases in India; legal base (Art 19(1)(c), RPA 29A, Symbols Order 1968, Tenth Schedule); State and national party criteria; benefits; national parties now; reforms (funding, bonds 2024, criminal records, internal democracy).

SCIENCE: Nuclear technology

Write: fission vs fusion; five reactor parts; reactor types; three stages; institutions; power plants; tests and doctrine; deals and laws; radioisotope uses; AP links (Kovvada, Tummalapalle, NFC, beach sands).